

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q13-0263

East Bismarck 100 kV
January 25, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0263. The Interconnection Customer proposes a 26.5 MW generating facility interconnecting at the East Bismarck 100 kV substation in the Illinois area. The request was received on August 21, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 182 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$262,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: LAKE JACKSON – EAST BISMARCK 100 KV
Contingency: E BISMARCK – TEXON 230 KV Line
Loading: 105.8% (Rating: 131 MVA)
DFAX: 25.8%
Pre-project loading: 100.5%
2. Facility: TUSCALOOSA T.S – COOPER NUCLEAR STA 100 KV Ckt 1
Contingency: TUSCALOOSA T.S – COOPER NUCLEAR STA 100 KV Circuit 2
Loading: 102.4% (Rating: 144 MVA)
DFAX: 6.7%
Pre-project loading: 101.2%
3. Facility: TUSCALOOSA T.S – COOPER NUCLEAR STA 100 KV Cir. 2
Contingency: TUSCALOOSA T.S – COOPER NUCLEAR SA 100 KV Ckt 1
Loading: 102.4% (Rating: 144 MVA)
DFAX: 6.7%
Pre-project loading: 101.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LAKE JACKSON – EAST BISMARCK 100 KV	Reconductor line	\$1,170,000	13% (\$153,000)
TUSCALOOSA T.S – COOPER NUCLEAR STA 100 KV	Reconductor line	\$681,000	8% (\$56,000)
TUSCALOOSA T.S – COOPER NUCLEAR STA 100 KV Line 2	Reconductor line	\$644,000	8% (\$53,000)
	Total allocated to project		\$262,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.